

EXP. NO: 17
PERFORMANCE EVALUATION OF DIGITAL COMMUNICATION
SYSTEMS USING EYE DIAGRAM ANALYSIS UNDER ADDITIVE

Objectives

1. To generate a digital communication signal in MATLAB and transmit it through an Additive White Gaussian Noise (AWGN) channel.
2. To obtain and analyze the eye diagram of the received digital signal for different noise conditions.
3. To measure eye-opening parameters such as eye height, eye width, crossing level and timing variation.
4. To study the effect of decreasing SNR / E_b/N_0 on inter-symbol interference, noise margin and timing performance.
5. To compare the transmitted and received signal characteristics and evaluate communication-system performance.

CODE

```
clc;
clear;
close all;

%% Parameters
N = 100;           % Number of bits
sps = 20;          % Samples per bit
SNR = 10;          % Signal-to-noise ratio (dB)

%% Generate Random Binary Data
data = randi([0 1],1,N);

%% NRZ Encoding (0 -> 0, 1 -> 1)
tx_signal = repelem(data,sps);

%% Pass Signal Through AWGN Channel
signalPower = mean(tx_signal.^2);
noisePower = signalPower / (10^(SNR/10));
noise = sqrt(noisePower)*randn(size(tx_signal));
rx_signal = tx_signal + noise;

%% Time Axis
t = (0:length(tx_signal)-1)/sps;
```

```

%% Plot Results
figure('Name','NRZ Encoding with AWGN','NumberTitle','off');

% Original Binary Data
subplot(4,1,1);
stairs(0:N-1,data,'LineWidth',1.5);
grid on;
title('Original Binary Data');
xlabel('Bit Number');
ylabel('Bit');
ylim([-0.2 1.2]);

% Transmitted NRZ Signal
subplot(4,1,2);
plot(t,tx_signal,'LineWidth',1.5);
grid on;
title('Transmitted NRZ Signal');
xlabel('Time (bit periods)');
ylabel('Amplitude');
ylim([-0.5 1.5]);

% Received Signal after AWGN
subplot(4,1,3);
plot(t,rx_signal,'r');
grid on;
title(['Received Signal with AWGN (SNR = ',num2str(SNR),' dB)']);
xlabel('Time (bit periods)');
ylabel('Amplitude');

% Eye Diagram (manual implementation)
subplot(4,1,4);
hold on;
segmentLength = 2*sps;

for k = 1:segmentLength:(length(rx_signal)-segmentLength)
    segment = rx_signal(k:k+segmentLength-1);
    timeSegment = (0:segmentLength-1)/sps;
    plot(timeSegment,segment,'b');
end

grid on;
title('Eye Diagram of Received Signal');
xlabel('Time (bit periods)');
ylabel('Amplitude');

```

```
xlim([0 2]);  
hold off;
```

```
%% Display Results
```

```
fprintf('Number of bits = %d\n',N);  
fprintf('Samples per bit = %d\n',sps);  
fprintf('SNR = %d dB\n',SNR);
```

OUTPUT

